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5.1 AC CIRCUITS - INTRODUCTION

An alternating voltage may be generated in two methods:

(i) By rotating a coil at constant angular velocity in a uniform magnetic field

(ii) By rotating a magnetic field constant angular velocity within a stationary coil

In either case the generated voltage will be of sinusoidal waveform. The magnitude of generated voltage depends upon:

(i) Number of turns of the coil
(ii) The strength of magnetic field
(iii) The speed of rotation

The first method is used for small A.C generators and the second method is used for large type of A.C generators.

In figure 5.1, an Alternating current shows the change in the direction of current and magnitude at regular intervals of power system. Alternating current plays a vital role in today's energy generation.

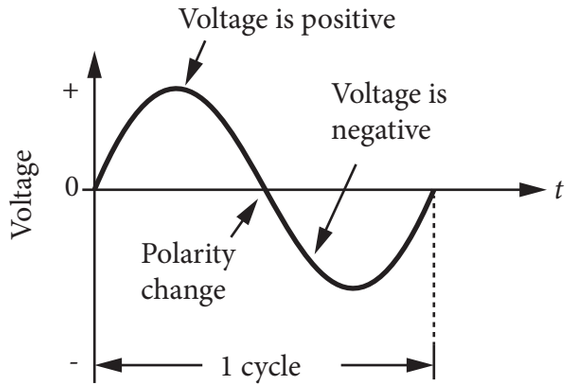


Fig 5.1 *Voltage vs Time*

5.2 AC WAVE FORM AND IT'S CHARACTERISTICS

A wave form is a representation of low alternating current (AC) that varies with time. The most familiar AC waveform is the sine wave, which derives its name from the fact that the current (or) voltage varies with the sine of the elapsed time.

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